

Evaluating the Impact of Multi-Agent Models on Stock Trading Decisions Compared to Single-Agent Models

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Abstract

Financial markets are usually unpredictable and full of uncertainty. The fundamental objective of most if not all investors is to maximise returns whilst minimising risks. Stock trading is one of the key methods that investors use to estimate the markets behaviour in order to ascertain the best possible action to take, with modelling being one of the main issues that investors face. This project investigates the application of a Multi-Agent System (MAS) to navigate the complexities of stock market trading by creating multiple decision-making agents. The system utilises three specialised agents: Technical, Health, and Sentiment. These agents evaluate different market factors with their outputs being aggregated into an Aggregation Agent, which employs weighted averages and conflict resolution logic to generate a final recommendation (Buy, Sell, or Hold).

Tested on Boeing, Apple, and Tesla over multiple years, the MAS was benchmarked against a stand-alone technical agent and a baseline buy-and-hold strategy. The simulations revealed that the system adapts the risk-avoiding strategy, showing robustness in simulations of “crash” years, such as incurring a return of 67.04 per cent during 2020 for Boeing Co, where the annual change of the stock’s price decreased by 34.40 per cent. The system also shows competitive performance during steady years, for example, during 2018 for Tesla Inc. where the system’s returns were 58.98 per cent. However, the system often fails to capitalise on the “boom” years. Although the system remains competitive in steady markets, its main strength lies in its risk-averse behaviour, demonstrating robustness in recognising patterns as-

sociated with market crashes and prioritising capital preservation. The findings highlight the efficacy of MAS in risk mitigation and suggest future improvements for dynamic agent weighting and broader data integration.

1 Introduction

Financial markets are defined by inherent volatility and complexity (1), as evidenced by the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic. Whilst the 2008 crisis stemmed from an internal collapse of the banking system driven by housing bubbles, the pandemic represented an external shock that triggered sudden market declines. In such environments, traditional trading methods often struggle to adapt to rapid changes. This has fuelled the demand for automated systems capable of processing large quantities of historical and real-time data.

This project explores the use of a Multi-Agent System (MAS) (2) to navigate financial complexity, focusing on the collaboration between distributed intelligent agents organised into three functional layers: a Data Preparation Layer, an Agent Layer, and an Aggregation and Plotting Layer.

2 Related work

The Multi-Agent System for Stock Trading (MASST) framework, proposed by Luo and Davis (3), addresses the increasing complexity of dynamic financial domains by implementing a multi-agent middle layer between the investor and distributed information sources. The architecture is designed to reduce

the workload of market analysis through three primary functional processes: automated stock information retrieval, continuous status monitoring and risk management, and a specialised decision support process for trading recommendations.

The MASST framework utilises nine distinct, interoperating agents to manage the lifecycle of an investment decision.

- **User Interaction Agents:** The Interface and Profiler agents manage user specifications and determine individual investor interests.
- **Operational Agents:** The Coordinator agent distributes tasks and plans execution, while the Monitoring and Communication agents track stock status and facilitate inter-agent data exchange.
- **Analysis Agents:** The Technical Analysis agent calculates traditional indicators and the Fundamental Analysis agent gathers macroeconomic data and market experts' opinions.
- **Executive Agents:** The Risk Management agent evaluates the safety of current holdings, while the Decision-Making agent combines the output of the analysis agents to generate buy, sell, or hold recommendations customised to the user's portfolio.

A critical aspect of this research is the emphasis on decision support rather than guaranteed profitability. The authors specify that the system aims to provide structural assistance in daily trading rather than a mechanism for "easy money." The framework was validated using data from the China Stock Market during a simulation period from June 2000 to February 2001, demonstrating that specialised agents can successfully operate simultaneously to collect, filter, and combine information from multiple sources to support investor workflows.

Technical indicators such as price and volatility indices remain central to modern trading decisions because they allow a technical agent to construct a multi-dimensional view of market behaviour. Within this framework, moving averages serve as a critical filter for identifying short-term trends by removing

stochastic noise that often obscures the directional movement of prices (4). The agent further refines this view by analysing the relationship between daily price change and daily price fluctuation; a high degree of fluctuation alongside a minimal net change often signals market indecision and increased risk. By combining volatility levels with volatility change, the system captures market sentiment, as increasing volatility is typically correlated with investor fear and potential trend reversals (5). These signals are validated through volume change, following Karpoff's (6) finding that a positive correlation between price and volume serves as a reliable buy signal, while a negative correlation identifies a weak move that is likely to fail.

The evaluation of a company's internal stability is equally vital and is based on well-established financial ratios derived from corporate reporting (7). The health agent leverages these metrics to determine whether a firm is expanding or contracting, using revenue growth and net income as primary indicators of financial strength. Although high costs may temporarily depress profit during expansionary phases, long-term stability is often predicted by profit margins, which measure a company's efficiency in converting sales into earnings. Companies with thin margins remain vulnerable to external shocks, such as increasing energy or material costs, which can quickly erode liquidity. To account for structural risk, the agent evaluates the debt-to-equity ratio and the current ratio, ensuring that a company is not overly dependent on external financing and has sufficient short-term assets to meet its obligations. Finally, the integration of earnings per share and the price-to-earnings ratio allows the agent to evaluate market valuation and determine whether a stock is overpriced relative to its earnings.

The financial ratios that assess the financial health of a company are well established (7). Key variables include growth profitability, leverage, liquidity, earnings performance, and market evaluation. These features are typically derived from financial statements, where financial information is engineered to determine a company's performance and stability. Knowing the financial 'health' of a company is key to making stock investment decisions, as it can be used to

assess risk, with high risk typically associated with companies that are financially unstable. However, the health of a company is not the only factor to consider in stock trading decisions, which is why the multi-agent system is important. For example, the price of a stock of a healthy company could be low/high, the sentiment could be positive or negative, and these factors as well as others should also be considered.

The Health agent used in this project uses multiple features to determine a company’s health, with the main variables being Profit margin and Earnings per share (EPS). The profit margin indicates how efficient the company can turn sales into profits. A higher profit margin suggests lower operational costs and better financial health with lower risks (8). Earnings per share (EPS) calculates how much profit is allocated to each outstanding share (9). These features, along with a couple of others that are later explained, makes the agent more robust. For example, a company may have strong earnings but may be in huge debt, or they may have low debt but lack profitability, or may be financially healthy but the stock could be overpriced. This combination of features allows for a better indicator of the health of the company.

FinBERT, used in the sentiment agent, is a transformer-based model. As the model is pre-trained on a large corpus of financial data, the model is therefore designed for financial terminology. It is used to identify the sentiment meaning of the news articles that relate to the companies. The result given from this model is a term Positive, Negative, or Neutral, as well as a confidence score for this result (10).

3 Methodology

3.1 System Overview

A MAS system was designed to make recommendations on stock trading decisions. The system is trained using historical data from the two years prior to the simulation year. When simulating, the agents generate daily predictions, combined by the aggregator and the plotting agent, to produce a final trading

decision for each day.

The system consists of three parts:

- **Data Layer:** Where financial and stock market data is collected, cleaned, and pre-processed into a format suitable for input into agent models.
- **Agent Layer:** Multiple, specialised agent models including the technical agent, health agent, and sentiment agent. Each agent is trained on a specific type of data that is relevant to the task of generating independent, weighted trade decisions with the output buy, sell, or hold.
- **Results Layer:** Where the aggregation model is trained to evaluate the weighted outputs from the three agents, which are then combined into one output for the final recommendation. The plotting agent puts the aggregation output results in a user-friendly format for user interpretability.

3.2 Data

Through the use of the yfinance package (11) in Python, Yahoo Finance’s (12) historical stock prices and financial information required for the technical and health agent are provided.

While yfinance provides the necessary data needed, the Alpha Vantage API is used specifically for the health agent’s data. It collects three financial components for a specific company, including the Balance Sheet, Income Statement, and Earnings Per Share (EPS). This source was utilised because yfinance cannot go back in history as far as the API in order to obtain the balance sheet, income statements, and EPS for each quarter.

However, the semantic agent had to use a completely different data set. A Kaggle dataset titled “Daily Financial News for 6000+ stocks”, which contained approximately 4 million articles across over 6000 stocks (13), had to be used instead of yfinance’s inbuilt headline storage, as yfinance only stored up to 10 news headlines per company, which means that any headlines before a certain date did not exist. This meant that the model would be restricted to

the near past, which ultimately did not allow for effective stress testing against tumultuous years like COVID-19.

3.3 Cleaning

Since Alpha Vantage returns data as strings with quarterly timestamps, the following preprocessing steps are performed:

- **Type Conversion:** All string values are cast to numerical data types.
- **Index Standardization:** Datetimes are converted to a uniform format.
- **Directional Merging:** As quarterly reports often fall on non-trading days such as weekends, data is joined to the yfinance dataset using a backward match where it joins to nearest preceding trading day.
- **Forward Fill:** To account for the disparity between quarterly reporting (4 entries/year) and active trading (≈ 251 days/year), a forward-fill method is applied to propagate fundamental metrics across all daily trading observations.

The sentiment analysis pipeline uses two distinct datasets sourced from Kaggle: Headlines, which contains article titles, publication dates, and source URLs, and News_Data, which includes company-specific descriptions and metadata. Due to significant data gaps within the News_Data file, a file merging step was performed on the common attribute URL.

A critical challenge identified during preprocessing was the inconsistency in temporal data, as timestamps were recorded across different timezones, specifically Eastern Standard Time (EST) and Eastern Daylight Time (EDT). To ensure a standardised approach for time-series analysis, a custom function, `parse_custom_date`, was used to convert all timestamps into a single denominator: Eastern Standard Time (EST).

In this context, actions represent the target variables for the model, categorised as buy, hold, or sell. Since the initial dataset lacked these labels, they were

generated using a `calculate_actions` function, which accepts the company ticker (e.g., AAPL) and the designated `simulation_year` as input.

The function retrieves stock data for the two years preceding the simulation year for training, and determines the optimal action by analysing the stock price five days into the future. The logic is defined as follows:

- **Buy:** If the price increases by at least 1 per cent
- **Sell:** If the price decreases by at least 1 per cent
- **Hold:** For all other price fluctuations

Each action $A \in \{1, 0, -1\}$ is further accompanied by a confidence score $C \in [0, 1]$. This score determines how confident that agent is in its predicted action.

The final five days are handled differently: the logic is applied relative to the very last available price in the training data, and the final day simply inherits the decision from the day prior. This labelling process was standardised across all agent training datasets.

Beyond target labels, the features for each agent were engineered based on established financial research. These variables were selected to effectively measure the fundamental health of a company, broader market sentiment, and technical price volatility.

3.4 Health Agent

3.4.1 Feature Engineering

As mentioned previously, the health agent processes several key financial ratios to quantify growth, profitability, and solvency:

- **Revenue Growth:** The percentage change in revenue between consecutive quarters. To ensure the first training day is populated, data from the two quarters prior to the training period is included.

$$\text{Revenue Growth} = \frac{\text{Revenue}_t - \text{Revenue}_{t-1}}{\text{Revenue}_{t-1}}$$

- **Net Income:** Extracted directly from the income statement, representing total profit after all deductions.
- **Net Profit Margin:** Measures the percentage of revenue remaining as profit:

$$NetProfitMargin = \frac{Net\ Profit}{Total\ Revenue}$$

- **Debt-to-Equity (DE) Ratio:** Evaluates financial leverage by comparing total liabilities to shareholder equity:

$$DE\ Ratio = \frac{Total\ Liabilities}{Total\ Shareholders'\ Equity}$$

- **Current Ratio:** A liquidity metric assessing the ability to cover short-term obligations:

$$Current\ Ratio = \frac{Current\ Assets}{Current\ Liabilities}$$

- **price to-Earnings (PE) Ratio:** The only daily-fluctuating fundamental feature that relates the market price to quarterly earnings:

$$PE\ Ratio = \frac{Market\ Value\ per\ Share}{Earnings\ per\ Share}$$

3.4.2 Modelling

The agent utilises a Random Forest Classifier with 500 estimators. A Training set comprising of the two year period preceding the simulation year before the model is ran.

The agent’s output is provided as a recommended action coupled with a confidence level. This is achieved through the `predict_proba()` function, which calculates the probability distribution across the three possible actions mentioned previously.

The final prediction is the action with the highest probability and the confidence level is the associating score provided along with that action.

3.5 Technical Agent

3.5.1 Feature Engineering

For the technical agent, several key features were engineered to capture price dynamics, market sentiment, and liquidity constraints:

- **10-Day Moving Average:** This represents the arithmetic mean of the closing price over a rolling 10-day window. The presence of null values in the initial nine days of the dataset was mitigated by using a backward-fill method, ensuring data continuity.
- **Daily Price Change (ΔP):** This metric captures the price movement of a day, defined as the difference between the daily opening price (P_{open}) and the closing price (P_{close}):

$$\Delta P = P_{close} - P_{open}$$

- **Daily Price Fluctuation (F_P):** This measures the total range of price movement during a single yfinance trading session, calculated as the difference between the daily high (P_{high}) and the daily low (P_{low}):

$$F_P = P_{high} - P_{low}$$

- **Volatility (VIX):** This feature utilises the closing value of the Volatility Index (VIX) from yfinance to serve as a proxy for “market fear.” It quantifies investor sentiment about risk and uncertainty, where higher values indicate greater market instability.
- **Volatility Change (ΔVIX):** Similar to price fluctuation, this represents the VIX range during a single yfinance trading session, calculated as the difference between the daily high (VIX_{high}) and the daily low (VIX_{low}):

$$\Delta VIX = VIX_{high} - VIX_{low}$$

- **Volume Change (ΔVol):** This represents the absolute change in the trading volume (number of shares traded) between the current day (t) and

the previous day ($t-1$). The value for the initial observation is set to zero:

$$\Delta Vol_t = Vol_t - Vol_{t-1}$$

3.5.2 Modelling

The agent first receives the data from the data_preparation class. Training sets, denoted X_{train} and Y_{train} , are then retrieved from the two year period preceding the simulation year.

As the technical agent operates under a supervised learning framework, the Random Forest algorithm was selected for classification. The ensemble was configured with $N = 500$ trees. This high number of estimators ensures the stability of the majority voting process and reduces variance in the predictions.

The agent’s output is provided as a recommended action coupled with a confidence level. This is achieved through the `predict_proba()` function, which calculates the probability distribution across the three possible actions mentioned previously.

The final prediction is the action with the highest probability and the confidence level is the associating score provided along with that action.

3.6 Semantic Agent

3.6.1 Feature Engineering

Within the merged dataset, the Text column contains a high volume of metadata encompassing multiple companies and international market news for any given day. To manage this density, the model utilises the `extract_target_sentences` function to isolate relevant information. This function performs a targeted keyword extraction, filtering the corpus to retain only those sentences containing the inputted company name or ticker. For example, if the target company is “Apple”, the function discards all unrelated noise and preserves only sentences that explicitly reference the firm. This preprocessing step is critical to ensure computational efficiency and to prevent the semantic agent from processing irrelevant data that could dilute the sentiment signal.

3.6.2 Modelling

The final component of the semantic agent is the `semantic_year` function, which generates a comprehensive semantic profile for a specific simulation year. To perform the sentiment analysis, the model utilises the FinBERT model. This specialised BERT model is pre-trained on a vast corpus of financial communication, and was selected due to its superior ability to contextualise domain-specific terminology (e.g., distinguishing “debt” or “volatile” in a financial sense rather than a general one), ensuring higher precision in sentiment classification.

The model processes target-specific sentences and generates a semantic value along with a confidence score. A significant challenge in this phase is the presence of null entries within the Text column for certain trading days. To maintain data continuity, the system implements a “look-back” mechanism:

- **Historical Propagation:** If a day lacks textual data, the model retrieves the most recent available sentiment value from the preceding 365 days.
- **Temporal Constraint:** The one-year window was selected as an upper bound, under the assumption that news older than 12 months loses its predictive relevance to current stock prices.
- **Neutral Default:** In instances where no textual data is available within the one-year look-back period, the function defaults the sentiment to a Neutral value with zero confidence.

3.7 Aggregation Agent

The aggregation agent serves as the decision maker of the system. Its primary objective is to corroborate the output of the technical, health, and sentiment agents into a single robust metric. This layer uses a normalised weighted average methodology to consolidate different market perspectives while allowing for varying levels of agent certainty.

3.7.1 Weighting

For each trading day, the agent retrieves the predicted actions $A \in \{1, 0, -1\}$ and their associated confidence scores $C \in [0, 1]$. To prioritise more reliable predictions, the system computes a confidence-weighted vote. The final normalised weighted average ($\bar{V}$) is formulated as:

$$\bar{V} = \frac{\sum_{i=1}^n (A_i \times C_i \times W_i)}{\sum_{i=1}^n (C_i \times W_i)}$$

where W_i represents a static weight assigned to agent i . While technical and health agents are assigned a full weight ($W = 1.0$), the sentiment agent is assigned a reduced weight of 0.8. This adjustment addresses two specific factors: the potential for repetitive data caused by the 365-day “look-back” mechanism and the inherently higher confidence scales produced by the FinBERT model, which could otherwise bias the final output.

3.7.2 Modelling

To transform the continuous average $\bar{V}$ into a discrete trading action, the agent applies an action threshold (τ). The default value is set to 0.2, though this remains changeable within the simulation environment for dynamic risk management:

- **Buy (1):** If $\bar{V} > \tau$
- **Sell (-1):** If $\bar{V} < -\tau$
- **Hold (0):** Otherwise

The aggregation agent incorporates safety refinements to handle high-variance market conditions where agents provide contradictory signals. A conflict is identified in two primary scenarios:

- **Triple Contradiction:** All three agents predict different actions
- **Dual Contradiction:** Technical and health agents provide opposing signals in the absence of a sentiment prediction.

If such a conflict occurs and the participating agents exhibit high confidence scores of > 0.5 , the system overrides the weighted average and defaults to a holding position. This ensures that the system does not execute a trade when the distributed intelligence is fundamentally deadlocked, prioritising capital preservation.

3.8 Plotting Agent

The plotting agent quantifies the success of the system by comparing its performance against benchmarks across several risk-adjusted dimensions.

Some metrics are first calculated:

- **Total Return:** This calculates the net monetary gain or loss as a percentage of the initial capital:

$$\text{Return} = \frac{\text{End Portfolio Value} - \text{Initial Capital}}{\text{Initial Capital}}$$

- **Maximum Drawdown (MDD):** MDD measures the largest peak-to-trough decline before a new peak is achieved, serving as a primary indicator of downside risk:

$$\text{Drawdown}_t = \frac{\text{Value}_t - \text{Peak Value}_{0\dots t}}{\text{Peak Value}_{0\dots t}}$$

The final MDD is the minimum value observed across the simulation period.

- **Annualised Sharpe Ratio:** This metric evaluates the excess return per unit of volatility. The numerator uses the 10-year CBOE Interest Rate as the risk-free rate (R_f), which is converted to a daily rate ($R_{f,daily} \approx R_{f,annual}/252$):

$$\text{Sharpe Ratio} = \left(\frac{\bar{R}_p - R_{f,daily}}{\sigma_p} \right) \times \sqrt{252}$$

where $\bar{R}_p$ is the mean daily percentage return and σ_p is the standard deviation of those returns.

From this, the agent produces three distinct visualisations:

- **Portfolio Time-Series:** A comparative line graph tracking the cumulative value of the MAS, the stand-alone agent, and the baseline (Buy and Hold) strategy over the simulation year.
- **Agent Vote Distribution:** A chart visualising the “Buy,” “Hold,” and “Sell” signals from the technical, health, and semantic agents to reveal behavioural patterns.
- **Confidence Profiles:** A time graph of each agent’s confidence scores, illustrating how certain the agents were during critical market shifts.

4 Results

The analysis of simulation results for all three independent agents and final aggregation agent is presented with an example of the company Apple (AAPL) over the year 2022. The stock prediction agent outputs the predicted probabilities for each buy, hold, and sell action based on the respective stock prediction criteria and then assigns the prediction action 1 for buy, 0 for hold, and -1 for sell based on the highest probability along with the associated confidence score for each available day over the simulation year dataset.

4.1 Health

Table 3 provides an analysis of Apple’s financial health throughout the year. The resulting prediction distribution strongly favours a “Buy” action for the majority of the period, recording 218 buy, 11 sell, and 22 hold signals. This concentration of buy recommendations suggests that the model perceived the company as financially robust and capable of sustained market performance throughout 2022.

The presence of 11 sell signals indicates that while the overall outlook was positive, the agent successfully identified specific critical intervals where fundamental indicators flagged potential risks. Furthermore, the limited number of hold decisions highlights high-conviction behaviour. Ultimately, the health agent signals a “safe betting” trend, underpinned by

Apple’s strong fiscal indicators and overall financial stability during the simulation period.

4.2 Technical

The data presented in Table 4 indicates a strong focus on price-action trends, suggesting a “Buy” signal for nearly half of the trading year—specifically 159 buy, 80 sell, and 12 hold predictions. This distribution demonstrates that the model successfully identified sustained upward market momentum, signalling optimal entry points for investment. Simultaneously, it maintained a defensive posture by flagging bearish periods, while the rarity of neutral “Hold” decisions points toward a high-conviction system. Ultimately, the technical agent favoured a bullish-biased trend for Apple in 2022, effectively balancing long-term growth windows with the identification of significant short-term downward shifts.

4.3 Semantic

Table 7 captures the market outlook based on news headlines, with the resulting distribution showing a strong tendency towards “Hold” actions. Specifically, the agent recorded 224 hold, 91 sell, and 50 buy signals. This high volume of neutral signals suggests that the model interpreted a significant portion of the news cycle as lacking strong directional indicators, favouring a “wait-and-see” approach.

For the remaining days, the model advised caution, identifying 91 sell actions in response to unfavourable or concerning news headlines. Conversely, the “Buy” signals were the least frequent, occurring on only 50 days when headlines exhibited significant positive “hype” or bullish sentiment. Overall, the sentiment agent reflects a widespread neutral trend, indicating that news coverage for Apple in 2022 was largely unattractive for aggressive trading, instead leaning toward caution and market observation.

4.4 Aggregation

In Table 8 on 3 January, all three agents advise to buy shares and the buy action is carried over to aggregated output with high confidence. Then on 4

January, both health and sentiment agents suggest buy action with good confidence in comparison with technical agent asking to hold with low confidence, where the weighted aggregation averages prediction to buying shares. At the end of the year, the agents exhibit conflict conditions with all three agents predicting different action with high confidence score (> 0.5). In this case, aggregation logic offers risk-control strategy of withholding shares to handle such strong disagreement scenarios. For the particular simulation use case, aggregation agent detects such 13 conflict conditions leading to a hold decision.

Table 9 indicates a combination of the three independent voting components. With 146 buy, 69 hold, and 36 sell predictions, the ensemble model effectively filters individual agent biases to produce a more balanced market outlook.

Although the Technical and Health agents were independently more aggressive, the Aggregation Agent moderated the final output. The increase in Hold signals (69) compared to individual agents suggests that the conflict resolution logic successfully identified periods of high market uncertainty where agents lacked a unified consensus. The prevalence of “Buy” actions confirms that the system identified sustained upward momentum for Apple in 2022, signalling optimal investment windows where financial health and technical indicators aligned. The 36 “Sell” decisions represent high-confidence bearish signals that survived the weighted average, providing a conservative exit strategy to protect capital during short-term downward trends.

In general, the aggregation agent effectively reconciles high-frequency technical signals with long-term financial health and news-based sentiment, resulting in a robust and high-conviction trading strategy for the simulation year.

4.5 Plotting

The simulation results, summarised in Table 10, assess the system under three distinct market conditions—steady, boom, and crash—for Boeing Co., Apple Inc., and Tesla Inc. These simulations utilise the technical agent in a stand-alone capacity and a

reduced-weight sentiment agent to evaluate the robustness of the MAS architecture.

The system demonstrates significant robustness during periods of high volatility and market decline. In the 2020 simulation for Boeing Co. (a crash year influenced by the 737MAX crisis), the MAS achieved a return of 67.04 per cent, dramatically outperforming the baseline return of -34.40 per cent. This success was driven by the system’s ability to aggregate a “Hold” signal from the sentiment agent, which reflected unclear news headlines—with technical triggers to avoid the catastrophic drawdowns seen in the stand-alone agent.

Similarly, for Apple Inc. in 2022 (crash year), MAS outperformed both benchmarks. Although the absolute return was negative (-21.68 per cent), it was superior to the -28.13 per cent loss from the baseline. This defensive capability is further evidenced by the superior Sharpe Ratio and lower MDD, confirming the system’s role as a capital preservation tool. In steady years, such as Tesla Inc. in 2018, the system yielded 58.98 per cent returns, successfully resolving conflicts between a bearish sentiment agent and bullish technical and health indicators.

Simulations for “boom” years, such as Boeing 2017 (+92.83 per cent baseline), Apple 2017 and Tesla 2020 (> 700 per cent annual growth), highlight the system’s inherent conservative bias. In these scenarios, the Health Agent frequently flagged rapid price appreciation as “dangerous” or “overpriced,” maintaining high confidence scores (> 0.8) for “Sell” or “Hold” recommendations.

Although this resulted in the MAS being significantly outperformed by the baseline approach in terms of total returns, the system maintained the lowest MDD in most boom simulations. This confirms that the MAS prioritises risk-averse behaviour over speculative profit-maximisation, acting defensively even as stock prices continue to rise.

5 Discussion

The results show that the MAS performed well during volatile and declining markets and consistently outperformed both the single agent and the buy-and-

hold baseline in “crash” scenarios, such as Boeing in 2020 and Apple in 2022 . This suggests that the aggregation of multiple signals allowed the system to identify and respond better to risk. These results aligned with the study conducted by Yong Luo (3) which showed that multi-agent models are designed to support decision-making under uncertainty rather than guarantee profitability. The purpose of the MAS is not to maximise returns, but to improve the quality and resilience of decisions by integrating multiple factors. However, the MAS developed by Yong Luo included more types of agents such as risk management and coordination layers, whereas the model developed in this study relies on only three agents. The simplicity of the model may limit the ability to capture complex market dynamics, and further development of more agents is required to factor in other variables that influence the stock trading decisions, such as the health of the overall market or competitors in the same industry.

The results suggest that the system behaves conservatively, which is demonstrated when comparing to the buy-and-hold baseline. The Buy-and-hold baseline is widely used as a benchmark in financial modelling because it represents a passive investment strategy that benefits from long-term market growth. However, it does not consider risk exposure or drawdowns, which means that outperforming or underperforming this baseline may not reflect the quality of a trading strategy in isolation. For example, although the MAS underperforms in terms of total return during high-growth periods, it consistently achieves lower maximum drawdown and improved Sharpe ratios in volatile conditions. This suggests that the MAS is more of a risk-adjusted strategy instead of a return-maximising one. Therefore, comparing with a buy-and-hold baseline highlights the conservative bias of the system, but it can also under-represent its value in risk-sensitive contexts.

The behaviour of individual agents further explains the overall performance of the system. The technical agent demonstrates strong trend follower capabilities, generating frequent “Buy” signals during upward price movements. This is consistent with the findings of Jonathan M. Karpoff, who identified a relationship between price movements and trading

volume as a useful indicator of market direction (6). However, the high frequency of signals also suggests a tendency towards over-trading, which may increase exposure to short-term noise. In contrast, the sentiment agent predominantly produces “Hold” decisions, reflecting the difficulty of extracting strong directional signals from financial news . Although the use of the FinBERT model enhances domain-specific sentiment analysis, the prevalence of neutral outputs suggests that news data alone may lack sufficient predictive power, particularly when filtered through a look-back mechanism. This limitation reduces the influence of sentiment within the aggregation process and can contribute to the overall conservatism of the system.

The health agent provides long-term insight into the overall financial health of a company. However, quarterly reports caused a delay that limited its ability to respond to rapid market changes. This aligns with research conducted by Stephen H. Penman, who demonstrated that financial ratios give an incomplete and delayed representation of company performance (7). The Tesla 2022 simulation demonstrates how training on outlying “boom” years can bias the model to make over optimistic predictions, which reduces the models ability to detect downturns in subsequent years. This is consistent with the challenges associated with learning models applied to financial data, which are sensitive to regime changes and historical bias. The MAS partially mitigates this through aggregation but more adaptive systems should include dynamic weighting.

The results illustrate the risk-return trade-off, and the MAS reduces volatility and drawdowns at the expense of lower returns in high-growth periods. This indicates that the model prioritises capital preservation over profit maximisation. This trade-off is shown by the high use of “Hold” decisions by the systems during periods of uncertainty, which prevents large losses but also limits gains. Although this behaviour is safe for investors, it reduces the competitiveness of the system against simpler strategies in growing markets.

In general, MAS outperforms single-agent approaches in terms of robustness, risk management, and considers more variables. Its performance aligns

with existing literature on multi-agent systems where stock market decisions are improved. However, compared to the buy-and-hold baseline, it revealed that these advantages are context-dependent because the MAS underperformed in conditions where risk-taking is rewarded. The findings suggest that although MAS systems have potential, their effectiveness depends on design choices and evaluation metrics.

Although the evaluation incorporates multiple performance metrics and market conditions, statistical significance testing was not conducted. This study focuses on deterministic historical simulations over a limited number of stocks and years. The results were driven by specific market conditions rather than repeated stochastic trials. A meaningful statistical analysis would require repeated simulations and the observed differences should be interpreted as indicative rather than conclusive.

5.1 Limitations

Whilst the study was a success, there are still some limitations relating to each agent and the data itself.

The main limitation of this study is the data. Although yfinance provides a solid data resource, the data needed for the Health Agent is not as detailed as the study needs it to be. Although Alpha Vantage provides a solid resource for the data of the Health agent, other API resources provide more detailed and rich financial information. Regarding the sentiment agent, the number of headlines is limited, as the distribution of stock headlines is not even. Some companies have far more entries in the dataset for certain years than the others.

A key limitation of the Health agent is that it mostly uses financial ratios and information from quarterly balance sheets and income statements. This means that values remain constant for extended periods, which makes it difficult to align with daily stock prices. This creates a lag between the real company performance on that day as it could represent figures from up to three months beforehand, except for the price-to-earnings ratio, which is reported daily. This limits the agents responsiveness to new information.

A key limitation of the Technical agent is that some

of the indicators used, such as MACD, relies on historical data. This can sometimes delay the Technical agent to enter or exit the market, resulting in missing the new trend or staying too long after the reversal. This can again be fixed by using a different metric or by using a different API.

A key limitation of the Semantic agent is the lack of data as mentioned previously. A news article on how the company is performing is not provided on a daily basis but instead is shown sporadically in the data set. Although the addition of a look-back function to take the closest headline to the day provided is given, there will still be some missing data. A way to fix this would be to increase the size of the dataset.

A key limitation of the Aggregation agent is limited by its dependency over choice of configurable parameters. While customisation offers flexibility for user preferences, it also makes the model sensitive to tuned configurations. The action threshold defines the mapping of weighted average into best action, and an inappropriate selection of this parameter can lead to misguided trading actions. The challenge of high confidence scores from the Semantic agent overpowering the other agents is addressed by using an input switch down-weighting the sentiment contribution for the entire simulation year, which correctly mitigates the impact of misleading headlines for the models duration. However, this can potentially lead to undervaluing sentiments in the duration when headlines are more valuable and relevant than statistical data to read market conditions.

A final limitation of the model is the choice of the training and testing years. The years chosen influence how well the model can perform with respect to anomalous data. The Tesla 2022 simulation reveals a critical vulnerability with respect to historical training dependencies. Because the system was trained on the anomalous “boom” years of 2020 and 2021, the health agent failed to recognise the 2022 crash, recommending “Buy” throughout the year. This suggests that the internal assessment of financial health can become skewed when the model is trained on periods of extreme, sustained growth followed by an immediate crash.

6 Conclusion

This study successfully developed and evaluated a Multi-Agent System designed to provide optimised stock investment advice by decentralising decision-making across three specialised domains: technical, health, and sentiment. Using a dynamic weight allocation protocol, the system effectively outputted various market signals into a coherent trading strategy.

The simulation results across Boeing Co., Apple Inc., and Tesla Inc. demonstrate the robust capabilities of the system. The MAS was particularly effective during “crash” years, such as Boeing in 2020 and Apple in 2022. In these scenarios, the conservative bias of the system acted as a safeguard, significantly minimising capital losses compared to baseline buy-and-hold strategies.

Although the system remained competitive during “steady” growth years (e.g., Boeing 2016, Apple 2016, and Tesla 2018), it demonstrated a clear limitation in “boom” years. During periods of aggressive price appreciation, the system’s risk-averse nature prevented it from fully capitalising on massive upward trends. This confirms that the current MAS architecture prioritises capital preservation and low-risk exposure over aggressive profit maximisation.

The project confirms that a multi-agent approach, where specialised agents focus on isolated factors before aggregation, significantly enhances the depth of financial modelling. By balancing price action, fundamental corporate health, and news-based sentiment, the MAS provides a holistic perspective that traditional single-agent models often lack. However, the inherent unpredictability of financial markets ensures that challenges remain in terms of scalability and predictive accuracy.

6.1 Future Work

To address the current limitations and further refine the MAS, several avenues for future research are proposed:

- **Training Window Calibration:** Investigating the optimal duration for historical training data to allow the system to recognise growth patterns

while avoiding the current “conservative bias” during times that hamper performance.

- **Data:** Incorporating more advanced datasets, such as real-time data or detailed macroeconomic indicators, to provide agents with a more comprehensive information set.
- **Advanced Sentiment Extraction:** Expanding the sentiment agent’s corpus to include diverse sources, such as social media platforms and earnings call transcripts.
- **Meta-Learning Aggregation:** Replacing the dynamic weight voting mechanism with a learner agent. This would allow the system to learn which specific agents are most reliable during particular market conditions (e.g., relying more on the health agent during earnings seasons or the sentiment agent during high-volatility events).

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7 Contribution

Task	Adam Balatka	Pangkaew Chansri	Emily Gomer	Pinal Lad	Torin Lindsay
Planning	20	20	20	20	20
Pre-processing	20	0	0	0	0
Technical Agent	0	20	0	0	0
Health Agent	0	0	20	0	0
Semantic Agent	0	0	0	0	20
Aggregation Agent	0	0	0	20	0
Report	20	20	20	20	20
Overall	20	20	20	20	20

8 Images and Tables

Table 2: Health Agent Simulation Results

Date	Sell Prob	Hold Prob	Buy Prob	Action	Conf.
2022-01-03	0.382	0.014	0.604	1	0.604
2022-01-04	0.478	0.012	0.510	1	0.510
2022-01-05	0.490	0.012	0.498	1	0.498
2022-01-06	0.480	0.022	0.498	1	0.498
2022-01-07	0.480	0.022	0.498	1	0.498
⋮	⋮	⋮	⋮	⋮	⋮
2022-12-23	0.112	0.064	0.824	1	0.824
2022-12-27	0.112	0.064	0.824	1	0.824
2022-12-28	0.112	0.064	0.824	1	0.824
2022-12-29	0.112	0.064	0.824	1	0.824
2022-12-30	0.112	0.064	0.824	1	0.824

Table 3: Health Agent Prediction Distribution

Predicted Action	Count
Buy (1)	218
Sell (-1)	11
Hold (0)	22

Table 4: Technical Agent Simulation Results

Date	Sell Prob	Hold Prob	Buy Prob	Action	Confidence
2022-01-03	0.340	0.300	0.360	1	0.360
2022-01-04	0.310	0.370	0.320	0	0.370
2022-01-05	0.258	0.234	0.508	1	0.508
2022-01-06	0.308	0.304	0.388	1	0.388
2022-01-07	0.388	0.166	0.446	1	0.446
...	...	...	...	...	...
2022-12-23	0.688	0.138	0.174	-1	0.688
2022-12-27	0.574	0.226	0.200	-1	0.574
2022-12-28	0.566	0.180	0.254	-1	0.566
2022-12-29	0.624	0.176	0.200	-1	0.624
2022-12-30	0.548	0.214	0.238	-1	0.548

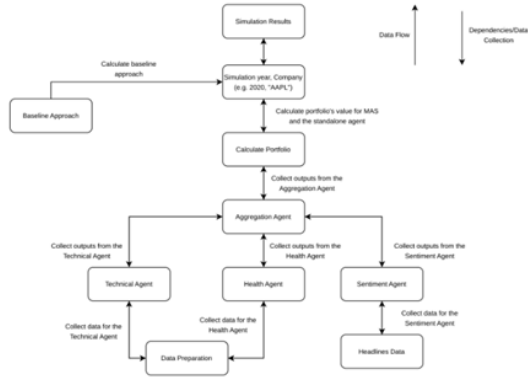


Figure 1: The Process undertaken from start to finish within the project

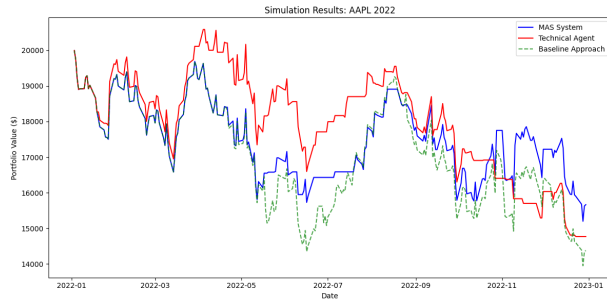


Figure 2: Simulation Results: AAPL 2022 Portfolio Value Comparison

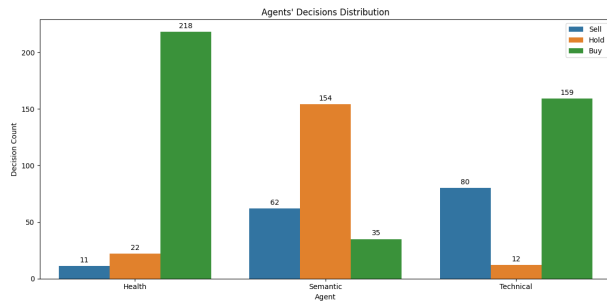


Figure 3: Distribution of Trading Decisions by Agent Type

Table 5: Technical Agent Prediction Distribution

Predicted Action	Count
Buy (1)	159
Sell (-1)	80
Hold (0)	12

Table 6: Sentiment Agent Simulation Data

Target Date	Headline	Label	Score	Pub. Date	Action
2022-01-01	Weekly Preview: Earnings to Watch This Week (AAPL)	positive	0.866	2021-10-10	1
2022-01-02	Weekly Preview: Earnings to Watch This Week (AAPL)	positive	0.866	2021-10-10	1
2022-01-03	Weekly Preview: Earnings to Watch This Week (AAPL)	positive	0.866	2021-10-10	1
2022-01-04	Weekly Preview: Earnings to Watch This Week (AAPL)	positive	0.866	2021-10-10	1
2022-01-05	Weekly Preview: Earnings to Watch This Week (AAPL)	positive	0.866	2021-10-10	1
⋮	⋮	⋮	⋮	⋮	⋮
2022-12-27	Implied MTUM Analyst Target Price: \$159	neutral	0.905	2022-08-30	0
2022-12-28	Implied MTUM Analyst Target Price: \$159	neutral	0.905	2022-08-30	0
2022-12-29	Implied MTUM Analyst Target Price: \$159	neutral	0.905	2022-08-30	0
2022-12-30	Implied MTUM Analyst Target Price: \$159	neutral	0.905	2022-08-30	0
2022-12-31	Implied MTUM Analyst Target Price: \$159	neutral	0.905	2022-08-30	0

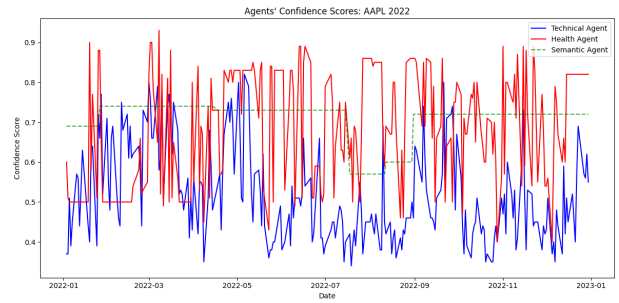


Figure 4: Time Series of Agents' Confidence Scores for AAPL 2022

Table 7: Sentiment Agent Prediction Distribution

Predicted Action	Count
Hold (0)	224
Sell (-1)	91
Buy (1)	50

Table 8: Aggregation Agent Simulation Results

Date	T-Act	T-Conf	H-Act	H-Conf	S-Act	S-Conf	W-Avg	Final	Decision
2022-01-03	1	0.36	1	0.60	1	0.87	1.000	1	BUY
2022-01-04	0	0.37	1	0.51	1	0.87	0.789	1	BUY
2022-01-05	1	0.51	1	0.50	1	0.87	1.000	1	BUY
2022-01-06	1	0.39	1	0.50	1	0.87	1.000	1	BUY
2022-01-07	1	0.45	1	0.50	1	0.87	1.000	1	BUY
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
2022-12-23	-1	0.69	1	0.82	0	0.91	0.054	0	HOLD
2022-12-27	-1	0.57	1	0.82	0	0.91	0.109	0	HOLD
2022-12-28	-1	0.57	1	0.82	0	0.91	0.109	0	HOLD
2022-12-29	-1	0.62	1	0.82	0	0.91	0.085	0	HOLD
2022-12-30	-1	0.55	1	0.82	0	0.91	0.118	0	HOLD

Table 9: Aggregation Agent Prediction Distribution

Decision	Count
BUY	146
HOLD	69
SELL	36

Table 10: Performance Comparison: MAS vs. Stand-alone Agent vs. Baseline (bold values show winner per metric)

Stock	Year	MAS Ret.	Agent Ret.	Base Ret.	MAS SR	Agent SR	Base SR	MAS MDD	Agent MDD	Base MDD
Boeing	2016	26.48%	-5.51%	14.48%	1.4299	-0.2844	0.6145	-11.40%	-22.32%	-22.32%
	2017	14.73%	17.43%	92.83%	1.8830	2.1789	3.7244	-1.19%	-1.19%	-4.50%
	2020	67.04%	-35.71%	-34.40%	1.0311	-0.0869	-0.0662	-37.32%	-72.23%	-72.08%
Apple	2016	8.14%	5.37%	12.37%	0.3808	0.2677	0.5403	-18.90%	-15.43%	-18.90%
	2017	17.68%	6.54%	47.99%	2.0597	1.4481	2.1953	-0.59%	0.00%	-8.85%
	2022	-21.68%	-26.17%	-28.13%	-0.6815	-1.0662	-0.8396	-24.02%	-28.28%	-30.27%
Tesla	2018	58.98%	-30.42%	4.16%	1.0800	-0.5873	0.3069	-22.38%	-38.71%	-33.96%
	2020	41.77%	-11.10%	707.14%	1.2241	-0.1526	2.7876	-18.83%	-44.06%	-60.62%
	2022	-69.04%	-47.02%	-69.10%	-1.5348	-1.3227	-1.5259	-72.56%	-54.02%	-72.71%